INDIAN INSTITUTE OF INFORMATION TECHNOLOGY RAICHUR
MATHEMATICS CLUB
PROBLEM NUMBER: WP2603
DATE: JANUARY 29, 2026
DUE DATE: FEBRUARY 11, 2026

WEEKLY PROBLEM

EXERCISE. A rare species of honeybee builds hives made of hexagonal cells. A complete hive consists of **37 hexagonal cells** arranged in concentric rings:

- Center: 1 hexagon
- 1st Ring: 6 hexagons
- 2nd Ring: 12 hexagons
- 3rd Ring: 18 hexagons

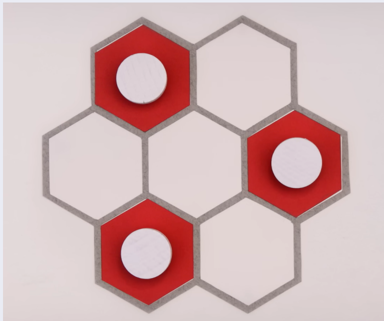

Figure 1: Hive Arrangement

Setup: You are given **60 bees** to build as many complete hives as possible based on the following rules:

1. **Direct Placement:** You may place a bee directly into a cell to fill it (consumes 1 bee).
2. **Automatic Filling:** Any empty cell with **3 or more filled neighbors** fills automatically (consumes 0 bees).
3. Automatic filling can trigger chain reactions.

Questions:

1. What is the **minimum** number of bees required to fill one complete hive of 37 hexagons?
2. Using this optimal strategy, what is the **maximum** number of complete hives that can be built using 60 bees?

(Scan this to submit solution!)